

Ziquan Wei

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EDUCATION

- University of North Carolina at Chapel Hill, USA *PhD in Computer Science, Aug 2023 - May 2028*
- Huazhong University of Science and Technology, China *Master in Biomedical Engineering, Sept 2019 - June 2022*
- Yunnan Normal University, China *Bachelor in Computer Science, Sept 2015 - June 2019*

EXPERIENCE

- University of North Carolina at Chapel Hill  *Research Assistant, Sept 2022 - Current*
 - **Medical Conditional Generative Models:** Proposed DiffDT to involve digital twins that are generated in conditions of EHR for disease reasoning. Developed Time2Track, an EMC foundation model, for ECG-to-4D-mesh generation [C1][C2].
 - **Brain Foundation Models:** Built BrainMoE using a novel cognition joint embedding. Developed Large Connectome Model 1.2B with a multitask supervised pretraining landscape to interpret brain-environment interactions [C3][C4][C6].
 - **Detour Pathway Transformers:** Discovered detour structure in brain network, leading to a novel Graph Transformer to uncover structural-functional coupling in human connectomes [J2][C5][C7].
 - **3D Medical Image Segmentation:** Engineered a high-throughput deep learning pipeline for 3D nuclei instance segmentation in teravoxel microscopy images [J3][C8].
 - **HPC Administration:** Administrator for a 10-node computing cluster. Manage 20 GPUs and over 1.5PB of storage.
- EGRA.ai  *Research intern, May 2026 – Aug 2026*
 - **Reinforcement Learning from Biofeedback:** Revise RLHF using additional brain signal rewards.
- Nanyang Technological University  *Research Intern, Jun 2021 – Sep 2021*
 - **Multimodal Emotion Recognition:** Successfully integrated audio, visual and EEG streams to improve recognition accuracy of human driver's emotion in dynamic environments [C9].

PROJECTS

- **CyberNeuro: Auto Statistics for Tabular Data + Agent skills for brain MRI Pre-processing** *Jan 2026 - May 2026*
Tools: [Ollama, FastMCP, TypeScript, Vite, Agent-to-agent, Agent tool calling]  Kaggle Writeup
 - **Lead the project.** Created tasks for the team and reviewed the project development.
 - **Created backend infrastructure** of MCP, Agent-to-Agent, tool visualization, and robust agent tool calling for an interactive and privacy-preserving workflow.
- **CellPheno: Segmenting and Classifying Cellular Resolution Teravoxel Microscopy Images** *Feb 2023 - Feb 2026*
Tools: [C++, Python, Nifti, Unet, GNN, nextflow]  bioRxiv,  nextflow
 - **Built high-throughput computation tools** for efficient and error-resistant teravoxel whole-brain LSFM image analysis workflow for segmentation, cell classification, visualization, and quality control in a large cohort ($n=118$).

FIRST-AUTHOR PUBLICATIONS

C=CONFERENCE J=JOURNAL * =IN SUBMISSION † =CO-FIRST AUTHOR

- C1. [NeurIPS '26]* [ZW](#), et al. **From Heartbeat to Cardiochoreography: A Mechanics Foundation Model for Individualized 4D Cardiac Motion Generation Conditioned on Electrophysiology** (ratings 5 4 4).
- C2. [ICML '26] [ZW](#), et al. **Marrying Generative Model of Healthcare Events with Digital Twin of Social Determinants of Health for Disease Reasoning.**
- J1. [bioRxiv]* [FAK](#)†, [IC](#)†, [ZW](#)†, et al. **Chd8 Haploinsufficiency Leads to Molecular Layer Heterotopias and Agedependent Cortical...**
- J2. [MedIA '26] [ZW](#), et al. **NeuroDetour: A Neural Pathway Transformer for Uncovering Structural-Functional Coupling...** (IF=11.8).
- J3. [ARBME '26] [ZW](#), et al. **Teravoxel Microscopy Image Analysis for Neurological Diseases.** (IF=9.6).
- C3. [AAAI '26] [ZW](#), et al. **Large Connectome Model: An fMRI Foundation Model of Brain Connectomes Empowered by...**
- C4. [NeurIPS '25] [ZW](#), et al. **BrainMoE: Cognition Joint Embedding via Mixture-of-Expert Towards Robust Brain Foundation Model.**
- C5. [NeurIPS '24] [ZW](#), et al. **NeuroPath: A Neural Pathway Transformer for Joining the Dots of Human Connectomes.**
- C6. [MICCAI '25] [ZW](#), et al. **Brain-Environment Cross-Attention (BECA) Meta-Matching: A New Perspective of Brain Connectome Zero-Shot Learning.** (EA, 9%)
- C7. [MICCAI '24] [ZW](#), et al. **Representing Functional Connectivity with Structural Detour: A New Perspective to Decipher Structure-Function Coupling Mechanism.**
- C8. [MICCAI '23] [ZW](#), et al. **A General Stitching Solution for Whole-Brain 3D Nuclei Instance Segmentation from Microscopy...** (EA, 14%)
- C9. [ICCVW '21] [SZ](#)†, [YD](#)†, [ZW](#)†, et al. **Continuous emotion recognition with audio-visual leader-follower attentive fusion.**